

[P5] Lower bounds for $g(r)$: $3r - 3$ for every r , and $3.1024r$ for large r

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Abstract

Sivashankar [2] proved $g(r) \geq 3r - 4$ for every r by deleting the cards of high-degree symbols and bounding the cover number of what remains, a family of maximum degree three, by $4r \leq q + r + 4$. He asked whether $+4$ can be replaced by $+3$. Theorem 1 gives a proposed proof that it can, so $g(r) \geq 3r - 3$ for every r . The residual constant $+3$ cannot be lowered [2], and $g(3) = 6$ and $g(4) = 9$ show that $3r - 3$ is attained.

Theorem 2 excludes the equality case for every $r \geq 16$, so $g(r) \geq 3r - 2$ there. Its proof uses three tools in turn: a count of matching numbers for large remainders, a swap lemma for the deleted cards, which rules out the rigid cases outright, and an anchor count for the remaining ones. At $r = 15$ one case survives all three, and §4.5 says why.

Section 5 turns the exclusion of equality into a quantitative statement (Theorem 3), with a gain linear in r when the remainder is large. Section 6 adds a second saving, taken from the deleted cards, that is largest when the remainder is small. It uses a bound on the deleted cards of the last steps of the peeling, a budget for the matching numbers of their traces, and a rainbow matching theorem of Correia, Pokrovskiy and Sudakov [5]. Together they give Theorem 4: $\liminf g(r)/r \geq 3 + (65 - 36\sqrt{2})/138 = 3.1020\dots$, above the constant $(41 - \sqrt{19})/12 \approx 3.0534$ of [2]. At the extreme point of that bound about half of the deleted cards have traces with small matching number. Such cards still carry many witnesses, and using them gives Theorem 5, $\liminf g(r)/r \geq 3.1024745$, where the final minimisation is certified by interval arithmetic.

The proofs are proposed and have not been independently reviewed. Every finite step is checked exhaustively by a script named in §7. No priority is claimed.

1. Setting

$g(r)$ is the least number of cards in a pairwise intersecting r -uniform family H with $\tau(H) = r$. The known values are $g(3) = 6$, $g(4) = 9$, $g(5) = 13$ [4], and $g(6) = 17$ ([P1]–[P4] of this set). Against $3r$ these are $3r - 3$, $3r - 3$, $3r - 2$ and $3r - 1$. So a bound $g(r) \geq 3r - c$ valid for every r cannot have $c < 3$; a bound $3r - 2$ can hold only from $r = 5$ on, and $3r - 1$ only from $r = 6$ on.

For large r , [2, §4] proves $g(r) \geq ((41 - \sqrt{19})/12 - o(1))r$, about $3.053r$, for r beyond a threshold that is not made explicit. Theorems 1 and 2 have coefficient 3 and hold with explicit constants at every r . Theorem 4 is asymptotic, like [2], and raises the coefficient to 3.1020 in closed form; Theorem 5 raises it to 3.1024.

What is borrowed. The peeling reduction, the dual triple system, the maximum matching, the selected witnesses, the first local support bound, and the sharpness family are from [2, §2–3]. They are recalled with proofs so that the argument can be read on its own.

Peeling. Delete the cards containing a symbol of current degree at least four, and repeat until no such symbol is left. If k symbols are chosen and the remainder J has q cards, then

$$|H| \geq q + 4k, \quad r \leq k + \tau(J),$$

because the k chosen symbols together with any cover of J cover H . Hence a bound $4\tau(J) \leq q + r + C$ gives $|H| \geq 3r - C$.

The dual view. For a symbol v of J let B_v be the set of cards containing it. Regard the cards as points. The blocks B_v have size at most three, cover every pair of points, and every point lies in exactly r blocks. A cover of J is a set of blocks meeting every point. For any card E ,

$$q - 1 \leq \sum_{v \in E} (\deg v - 1) \leq 2r, \quad (1.1)$$

so $q \leq 2r + 1$.

Take a maximum matching $M_1, \dots, M_t$ among the distinct three-element blocks. Let C be its union and U the other u points, so $q = 3t + u$; no three-element block lies inside U . A defining symbol of each M_i covers C , and pairing the points of U gives

$$\tau(J) \leq t + \lceil u/2 \rceil. \quad (1.2)$$

For each pair x, y of U choose one symbol w_{xy} in both, the *witness* of the pair. Witnesses are distinct, since a repeated one would lie in four cards or in a triple inside U . Each witness block is $\{x, y\}$ or $\{x, y, z\}$ with $z \in C$.

For each i let F_i be the *local graph* on U whose edge xy is present, with colour z , when the witness block is $\{x, y, z\}$ and $z \in M_i$. The F_i are simple and edge-disjoint. Two disjoint edges of one F_i never have different colours: their blocks would replace M_i by two disjoint triples. For each colour write e for its number of edges and s for its support, and let S_i be the sum of the three supports, $S = \sum_i S_i$.

2. Theorem 1: the sharp residual bound

Theorem 1 (proposed). *Every intersecting r -uniform family J with q cards and maximum degree at most three satisfies $4\tau(J) \leq q + r + 3$.*

Corollary 1. $g(r) \geq 3r - 3$ for every $r \geq 1$.

Proof. Peeling and Theorem 1. $\square$

2.1 Three inequalities

Capacity at U . The witnesses use exactly $u(u-1)$ incidences on U . The other $ur - u(u-1)$ incidences each reach at most two cards of C , and all $3tu$ pairs between U and C must be met. Hence

$$3tu - S \leq 2(ur - u(u-1)). \quad (2.1)$$

Capacity of a colour. Fix $z \in M_i$, and let the edges of colour z in F_i be e in number, with support s . In the card z , the e witnesses cover s points of U , a defining symbol of M_i covers its other two points, and the remaining $q - 3 - s$ points need the other $r - e - 1$ symbols, at most two each. So

$$2e - s \leq 2r - q + 1 =: d. \quad (2.2)$$

Local support ([2] for the first bound).

Lemma 1. *Let F be a simple graph on u vertices with at most three colours, in which disjoint edges always have the same colour. Then $S_F \leq \max(12, u + 4)$. If every colour has $2e - s \leq d$, then $S_F \leq \max(12, u + 2, d + 8)$.*

Proof. Suppose a colour has two disjoint edges, on four vertices W . Every other-colour edge lies in W . If no other colour is present, $S_F \leq u$. If the other-colour edges have no common vertex, every edge of the first colour meets all of them and lies in W too, giving 12. Otherwise at most two other-colour edges exist, sharing a vertex, with support contribution at most four, giving $u + 4$. Fix one of them, g . Every first-colour edge meets g , so $e \geq s - 2$, and $2e - s \leq d$ gives $s \leq d + 4$ and $S_F \leq d + 8$.

If every colour is internally intersecting, each is a star or part of a triangle. With all supports at most four, the sum is at most 12. Otherwise one colour is a star with at least four leaves, every other-colour edge contains its centre, and simplicity gives $S_F \leq u + 2$. $\square$

2.2 The preliminary inequality

Lemma 2 ([2]). $r \geq u + t - 2$.

Proof. Put $\alpha = r - u - t + 2$. If $u = 0$, (1.1) gives $\alpha \geq (r + 5)/3 > 0$. Otherwise (1.1), (2.1) and Lemma 1, with $K = \max(12, u + 4)$, give $t \leq u - 3 + 2\alpha$ and $\alpha \geq 1 - (K - u)t/(2u)$. A negative integer α would give $t \leq u - 5$ and then $\alpha > -1$: for $K = u + 4$ the bound is $-1 + 10/u$, and for $K = 12$ with $u \leq 8$ the needed strict inequality is $u^2 - 13u + 60 > 0$, whose discriminant is negative. $\square$

2.3 The odd equality case

By (1.2) and Lemma 2, if u is even then $4\tau \leq 4t + 2u = q + (t + u) \leq q + r + 2$. If u is odd, $4\tau \leq q + (t + u) + 2$. So Theorem 1 follows once *u odd implies $r \geq u + t - 1$* .

Suppose u is odd and, by Lemma 2, $r = u + t - 2$. Then $d = u - t - 3 \geq 0$, and (2.1) becomes

$$S \geq u(t + 2). \quad (2.3)$$

If $u \geq 11$, (2.3) and Lemma 1 give $t \geq u/2$. Then $d + 8 \leq u + 2$ and $12 \leq u + 2$, so every $S_i \leq u + 2$ and $u(t + 2) \leq t(u + 2)$, that is $u \leq t$, against $t \leq u - 3$.

For odd $u \leq 9$, the conditions $0 \leq t \leq u - 3$ and $u(t + 2) \leq t \max(12, u + 4)$ leave exactly

$$(u, t) \in \{(5, 2), (7, 3), (7, 4), (9, 5), (9, 6)\},$$

with $d = 0, 1, 0, 1, 0$ respectively.

Lemma 3 (small excess). *If every colour has $2e - s \leq 1$ and F has more than one colour, then $S_F \leq 12$, $S_F \neq 11$, and $S_F = 12$ only when F is a properly three-coloured K_4 . If also $u \leq 9$, the same holds when F has one colour.*

Proof. A connected component on $v \geq 2$ vertices has $2e - v \geq v - 2$, so each colour is a matching, or one path on three vertices plus a matching. A colour with a matching of size three leaves the other colours empty, so F has one colour and $S_F \leq u \leq 9$. A colour of support five is P_3 plus an edge, and the other colours are then confined to two edges, giving at most 9. Otherwise every support is at most four. Support four forces two disjoint edges, and two cross-intersecting two-edge matchings are two perfect matchings of one K_4 ; a third colour must be the third. $\square$

The cases $d = 1$. For $(9, 5)$, (2.3) asks $S \geq 63$ and Lemma 3 allows 60. For $(7, 3)$, (2.3) asks 35, so all three S_i are 12: three edge-disjoint copies of K_4 on seven vertices. Two such copies meet in one vertex and leave $K_{3,3}$ plus an isolated vertex, which holds no third K_4 .

The cases $d = 0$. Now every colour is a matching and $q = 2r + 1$, so (1.1) is an equality for every card: every symbol has degree three and any two cards meet once. So every witness block is a triple and the F_i partition the edges of K_u . For $(5, 2)$, $qr = 55$ is not divisible by three. For $u \leq 9$ a local graph has at most six edges: one colour is a matching of at most four edges, and with several colours each is a matching of at most two, since a third disjoint edge would meet no other-colour edge. Six edges occur only as the three perfect matchings of a K_4 , and five only as K_4 minus an edge. For $(9, 6)$ the 36 edges of K_9 would split into six copies of K_4 , needing degree 8 divisible by three. For $(7, 4)$ the 21 edges of K_7 split into four parts of size at most six only as $6 + 6 + 6 + 3$, $6 + 6 + 5 + 4$ or $6 + 5 + 5 + 5$; two edge-disjoint K_4 leave $K_{3,3}$ plus a vertex, which excludes the first two. In the third, after one K_4 on four vertices, each copy of K_4 minus an edge contains a triangle and so takes one of the three edges among the other three vertices; it is then two of the four joined to two of the three, plus that edge. The three pairs among the three are distinct and meet, so edge-disjointness needs three disjoint pairs among four vertices, which is impossible.

This proves Theorem 1. $\square$

2.4 Sharpness, and degree four

For odd n take points $1, \dots, n$ and one symbol for each pair $([2])$. Then $q = n$, $r = n - 1$, a cover is an edge cover of K_n , and $\tau = (n + 1)/2$, so $4\tau = q + r + 3$.

Corollary 2. *If J has maximum degree at most four, then $4\tau(J) \leq q + r + 3$.*

Proof. Take a maximal matching of a four-element blocks; their defining symbols cover $4a$ cards and the rest has maximum degree three. $\tau \leq a + (q - 4a + r + 3)/4$. $\square$

Peeling at degree five then gives $|H| \geq (15r - q - 15)/4$ for the size q of the remainder, which improves the additive term in the corresponding step of [2] but not its coefficient.

3. The equality case

From here to §4, suppose $|H| = 3r - 3$ and $\tau(H) = r$ with $r \geq 5$.

3.1 The parameters

If H had a symbol of degree at least five, peeling it first would give $|H| \geq |J| + 4(r - \tau(J)) + 1 \geq 3r - 2$ by Theorem 1. If every degree were at most three, $3r - 3 \leq 2r + 1$ would force $r \leq 4$. So the

maximum degree is four. (At $r = 3$ and $r = 4$ the bound $3r - 3$ is attained, and this is where the argument stops.)

Take a maximum matching of a four-element blocks. Their $4a$ cards form a *groups* of four, and each group has its *group symbol*. Delete them and let J be the rest, of maximum degree three and $q = 3r - 3 - 4a$ cards. The a group symbols and a cover of J cover H , so $\tau(J) \geq r - a$, while Theorem 1 gives $4\tau(J) \leq q + r + 3 = 4(r - a)$. So Theorem 1 is an equality for J , which by §2.3 forces u odd, $\tau(J) = r - a$, $r = u + t - 1$ and $q = 3t + u$. Consequently

$$u = 2a + 3, \quad t = r - 2a - 2, \quad d = u - t - 1 \geq 0, \quad r = 2u - 2 - d, \quad (3.1)$$

and (2.1) becomes $S \geq ut$. Write $H_0 = (u - 1)/2$.

The number $d = 2r - q + 1$ measures the slack in (1.1). When $d = 0$ every symbol of a card of J has degree three in J ; as d grows, symbols of degree two and one are allowed.

3.2 Local bounds

Lemma 4 (pointwise capacity). *For $x \in U$ let S_x be the number of cards of C met by the witnesses at x . Then $S_x \geq t$. In one F_i , if x carries E edges in k colours, then $E - k \leq d$.*

Proof. The card x has exactly $u - 1$ witnesses, so exactly $r - (u - 1) = t$ other symbols. They meet at most $2t$ cards of C , and all $3t$ must be met, so $3t - S_x \leq 2t$. The $u - 1$ witnesses at x reach S_x distinct cards of C , so the number of witnesses “wasted” at x is $u - 1 - S_x \leq u - 1 - t = d$. Edges of one colour at x reach one card, so E edges in k colours waste $E - k$. $\square$

Lemma 5. *Every F_i with more than one colour has $S_i \leq \max(12, d + 8)$ and $S_i - e(F_i) \leq 6$. A one-coloured F_i has $S_i - e(F_i) \leq H_0$.*

Proof. The first case of Lemma 1 already gives $\max(12, d + 8)$. In the star case, F_i is a star at one vertex with E edges in $k \leq 3$ colours, and $S_i = E + k \leq d + 2k \leq d + 6$ by Lemma 4. For the second bound: with two disjoint edges of one colour on W , either every support lies in W and each colour has $s - e \leq 2$, or at most two other-colour edges exist, contributing at most two, while the first colour has $e \geq s - 2$. Otherwise every colour is a star or triangle with $s - e \leq 1$. A one-coloured graph has $s - e$ at most its number of components, at most H_0 . $\square$

3.3 Two augmentation bounds

Call a one-coloured F_i *large* if its matching number is at least four, and let m be their number.

Lemma 6. $2m \leq t$.

Proof. A large F_i uses one colour z_i ; its other two points of M_i are met by no witness. So every $x \in U$ must meet those $2m$ points with its t non-witness symbols. No such symbol meets two of them. Within one M_i , its triple and an edge of F_i avoiding x would replace M_i by two triples. Across M_i and M_j , an edge of F_i avoiding x and an edge of F_j avoiding x and those two endpoints, which matching number four provides, give three triples in place of two. Either way the matching was not maximum. $\square$

Lemma 7. $u \leq \max(12, d + 8)$.

Proof. Suppose not. By Lemma 5 every F_i with more than one colour has $S_i < u$, and a one-coloured F_i has $S_i \leq u$. Since $S \geq ut$, every F_i is one-coloured and spans U . Then $2e - u \leq d$ gives

$e \leq (u + d)/2$, and its matching number is at least $u - e \geq (u - d)/2 > 4$. So $m = t$, and Lemma 6 forces $t = 0$, whereas $t = u - d - 1 \geq 8$. $\square$

3.4 The swap lemma

The groups come from a *maximum* matching of four-element blocks. That alone constrains every deleted card.

Lemma 8 (swap). *Let $A \neq A'$ be deleted cards of the same group. If $w \in A$ and $w' \in A'$ are symbols of degree three in J , their sets of cards of J meet.*

Proof. w lies in A and in three cards of J , so it has degree four in H and its block is $\{A\} \cup T$ with T three cards of J ; likewise $\{A'\} \cup T'$. Neither block meets another group. If $T \cap T' = \emptyset$, the two blocks are disjoint, and putting them in place of the group's own block gives $a + 1$ disjoint four-element blocks, against maximality. $\square$

Call such T a *triple* of A . Every symbol that a deleted card shares with a card of J is a symbol of J , so a deleted card meets J through triples and through symbols of degree two or one in J .

Proposition 1. *In the equality setting $d \neq 0$, and $d \neq 1$ when $r \geq 11$.*

Proof. Take a group; it has four cards.

$d = 0$. By §2.3, $q = 2r + 1$ and every symbol of a card of J has degree three in J . So a deleted card A meets the $2r + 1$ cards of J through at most $r - 1$ triples. Fix one, T . If the other at most $r - 2$ all met T , they would reach at most $3 + 2(r - 2) = 2r - 1$ cards; so A has a triple T' disjoint from T . By Lemma 8 every triple of a group-mate A' meets both T and T' , so reaches at most one of the $2r - 5$ cards outside them. With at most $r - 1$ triples, A' misses one, since $r - 1 < 2r - 5$ for $r \geq 5$.

$d = 1$. Now $q = 2r$, and (1.1) allows each card of J at most one symbol of degree two and none of degree one. So J has at most $q/2 = r$ symbols of degree two, each in at most two deleted cards, since degrees in H are at most four: at most $2r$ incidences with deleted cards. A deleted card whose triples pairwise meet reaches at most $1 + 2n_3$ cards through n_3 triples and two through each other symbol, at most $1 + 2(r - 1) < 2r$ in all; so every deleted card has two disjoint triples. For a group-mate A' with n_2 symbols of degree two, the $2r - 6$ cards outside those two triples are reached at most once per triple and twice per symbol of degree two, so $2r - 6 \leq (r - 1) + n_2$ and $n_2 \geq r - 5$. Every deleted card is such a group-mate, so the $4a \geq 4$ deleted cards need at least $4(r - 5)$ incidences, more than $2r$ when $r \geq 11$. $\square$

4. Theorem 2: excluding equality from $r = 16$

Theorem 2 (proposed). $g(r) \geq 3r - 2$ for every $r \geq 16$.

4.1 The deleted cards and their matching numbers

For each of the $4a$ deleted cards A let L_A be the graph on U whose edge xy is present when $w_{xy} \in A$; write e_A , s_A , ν_A for its edges, support and matching number. Let $T = \sum_i e(F_i)$ and $D = S - T$.

A witness of degree two in J lies in at most two deleted cards, one of degree three in at most one, since degrees in H are at most four; so $\sum_A e_A \leq u(u - 1) - T$. Let $M = \sum_A (u - s_A)$. For $x \in U$ its t non-witness symbols have room for at most $3t$ other cards of H ; they must meet the $3t - S_x$ cards of C not met by witnesses, and every deleted card A whose graph misses x . So at most S_x deleted cards miss x , and $M \leq S$. Using $\nu_A \geq s_A - e_A$ and $4a = 2u - 6$,

$$\sum_A \nu_A \geq u^2 - 5u - D. \quad (4.1)$$

Lemma 9. *Within one group there are no three disjoint edges from three different graphs L_A .*

Proof. Extend them to a matching of H_0 edges of the complete graph on U , leaving one point x . The witnesses of these edges cover U except x and meet three cards of the group; one symbol of x that lies in the fourth card covers both. With the t defining symbols of C and the $a - 1$ group symbols of the other groups this is a cover of H of size $H_0 + 1 + t + a - 1 = r - 1$. $\square$

Lemma 10. *Three graphs each with a matching of size four and a fourth nonempty graph contain three disjoint edges from three different graphs.*

Lemma 11. *The same holds for four graphs with matching numbers at least 3, 3, 4, 4.*

Proof of both. Fix size-four matchings P, Q in two of the graphs and suppose no such triple exists. Call an edge e of another graph *blocked* if no two disjoint edges, one from P and one from Q , avoid it; every edge of the other graphs is blocked. The edges of P and of Q avoiding a blocked edge must cross-intersect. Each side keeps at least two, and one edge cannot meet three disjoint edges, so exactly two survive on each side and they form a four-cycle, which is a component of $P \cup Q$, with both ends of e in both supports. The other graphs supply three disjoint blocked edges (a size-three matching for Lemma 11, a size-four matching for Lemma 10). Their six endpoints lie in both supports, and outside one such four-cycle there are at most four such vertices, so some of them touches it and yields a second four-cycle. So $P \cup Q$ is two disjoint four-cycles on the same eight vertices, and every blocked edge is a diagonal of one of them.

For Lemma 11 the two size-three matchings each consist of three of the four diagonals. The first has both diagonals of one cycle and the second at least one of them; take that one for the second, the other for the first, and an edge of P in the other cycle. For Lemma 10 the three size-four matchings are the three perfect matchings of each of the two copies of K_4 . Take an edge f of the fourth graph. If f removes two vertices of one copy, the edge on its other two vertices and an edge of a different matching in the other copy avoid f . If f removes at most one vertex of each copy, each copy keeps a triangle or all of K_4 , which contains all three matchings, so two edges of different matchings avoid f . In every case a rainbow triple exists. $\square$

Order the matching numbers of one group as $v_1 \leq v_2 \leq v_3 \leq v_4$. Greedy choice gives a rainbow triple when $v_2 \geq 1$, $v_3 \geq 3$ and $v_4 \geq 5$. With Lemmas 9–11, every group has $v_2 = 0$, or $v_3 \leq 2$, or all $v_i \leq 4$ with neither $v_1 \geq 1$ and $v_2 = v_3 = v_4 = 4$ nor $v_1, v_2 \geq 3$ and $v_3 = v_4 = 4$. So the sum is at most $\max(2H_0, H_0 + 6, 13)$, and with (4.1):

$$D \geq \frac{u^2 - 6u - 3}{2} \quad (u \geq 15), \quad D \geq 39 \quad (u = 13). \quad (4.2)$$

4.2 Large remainders: u is at most 11

$u \geq 15$. By Lemma 7, $d \geq u - 8$ and $t = u - d - 1 \leq 7$. By Lemma 5, $D \leq 7H_0 = 7(u - 1)/2$. With (4.2) this gives $u^2 - 13u + 4 \leq 0$, false for $u \geq 13$.

$u = 13$. Lemma 7 gives $d \geq 5$. If $d \geq 6$ then $t \leq 6$ and $D \leq 6t \leq 36$, against (4.2). If $d = 5$ then $t = 7$ and every $S_i \leq 13$, so $S \geq 91$ forces every $S_i = 13$. A one-coloured F_i then has $e \leq 9$ and matching number at least four, so it is large and $S_i - e \leq 6$. A multicoloured F_i cannot be the

star case of Lemma 5, which gives at most $d + 6 = 11$, so $S_i = 13 = d + 8$ is the extremal case of Lemma 1: two other-colour edges pa , pb , and a first colour made of ab and six edges at p , so $S_i - e = 13 - 9 = 4$. Lemma 6 gives $m \leq 3$, so $D \leq 6 \cdot 3 + 4 \cdot 4 = 34$, against (4.2).

So $u \leq 11$.

4.3 $u = 11$: the anchor count

Call (x, c) *open* in a local graph F_j if no edge of colour c meets x and every edge of the other two colours does. Let A_j be the number of open pairs of F_j .

For a local graph F_i and $x \in U$, let P_x be the set of colours of F_i with no edge at x ; their points of M_i are not reached by the witnesses at x . Call a pair of colours in P_x *forbidden* if F_i has an edge of the third colour avoiding x , and call $c \in P_x$ *good* if, for every pair f of U that avoids x and is not an edge of F_i , some edge of F_i of a colour other than c avoids both x and f . For a set $R \subseteq P_x$ whose pairs are all forbidden, put g_x equal to the size of R , minus the number of colours of P_x outside R , minus the number of colours of R that are not good; take the best R , and let $G = \sum_x \max(g_x, 0)$, the *anchor value* of F_i .

Lemma 12 (demand). *If F_i has anchor value G , the other local graphs have at least $tu + G - S$ open pairs in all.*

Proof. Fix x and R . The t non-witness symbols of x must reach the $3t - S_x$ cards of C that its witnesses miss, among them the points of R . No symbol reaches two points of R : its block and an edge of the third colour avoiding x would replace M_i by two triples. So $|R|$ distinct symbols reach the points of R , each reaching at most one further card, and the other $t - |R|$ reach at most two each. The $|R|$ symbols therefore reach at least $t + |R| - S_x$ further cards missed by the witnesses, of which at most one per colour of P_x outside R lies in M_i .

Let $\{x, c, q'\}$ be such a symbol with $c \in R$ good and $q' \in M_j$, $j \neq i$. No edge of colour q' in F_j meets x , since the witnesses at x miss q' . If an edge f of F_j of another colour avoided x , goodness gives an edge e of F_i of a colour other than c avoiding x and f , and the blocks of $\{x, c, q'\}$, e and f are three disjoint triples in place of M_i and M_j . So (x, q') is open in F_j . Distinct q' give distinct pairs, and each colour of R that is not good spoils at most one. So at least $t + g_x - S_x$ open pairs come from x . Summing over the x with $g_x > 0$ and using $S_x \geq t$ for the rest gives $tu + G - S$. $\square$

Three anchors matter. A properly three-coloured K_4 on W has $G = 3(u - 4)$: for $x \notin W$ all three colours are missing, every pair is forbidden, and every colour is good, since the other two colours form a four-cycle on W and a pair f outside F_i meets W in at most one point. A spanning one-coloured large F_i has $G = 2u$: for every x its two unused colours are missing, forbidden together, and good, since a matching of size four leaves an edge avoiding x and f . The graph Y_k , with edges pa and pb of two colours and a third colour made of ab and k edges at p , has $G = 2(u - 3)$.

Lemma 13 (finite check). *Let $u = 11$ and $1 \leq d \leq 4$. Every local graph with more than one colour has at most $12 - S_j$ open pairs, and every one with $S_j \geq 11$ has anchor value at least 16. A one-coloured graph of support s has $11 - s$ open pairs, plus two for each vertex common to all its edges.*

The one-coloured count is direct. The rest is checked over every multicoloured local graph on eleven vertices that satisfies (2.2) and Lemma 4: those with $S_j \geq 11$ are the K_4 and Y_4 , and at $d = 4$ also Y_5 , with anchor values 21, 16 and 16.

Proposition 2. *In the equality setting, $u = 11$ with $1 \leq d \leq 4$ is impossible.*

Proof. Here $t = 10 - d$, every $S_j \leq 12$ by Lemma 5, and $S \geq 11t$. So $\sum_j (12 - S_j) \leq t \leq 9$ and some $S_j \geq 11$. That F_j is an anchor with $G \geq 16$: by Lemma 13 if it has several colours, and otherwise it spans U with at most $(11 + d)/2 \leq 7$ edges, hence matching number at least four and $G = 22$. By Lemma 12 the other local graphs carry at least $11t + 16 - S$ open pairs.

By Lemma 13 a local graph carries at most $12 - S_j$ open pairs, except a one-coloured one whose edges share a vertex. A star with k leaves has $12 - S_j = 11 - k$ and carries one more; a single edge or an empty graph would need $12 - S_j \geq 10 > t$. A star needs $11 - k \leq t$ and, by (2.2), $k \leq d + 1$, so $k = d + 1$ and it uses the whole budget; at most one occurs. So the local graphs carry at most $12t - S + 1$ open pairs, and $11t + 16 \leq 12t + 1$ would need $t \geq 15$. $\square$

4.4 Proof of Theorem 2

Suppose $r \geq 16$ and $|H| = 3r - 3$ with $\tau(H) = r$. By §4.2, $u \leq 11$, and $r = 2u - 2 - d$. If $u \leq 9$ then $r \leq 16 - d$, so $r = 16$, $u = 9$ and $d = 0$, excluded by Proposition 1. If $u = 11$ then $d = 20 - r \leq 4$: $d = 0$ is excluded by Proposition 1 and $1 \leq d \leq 4$ by Proposition 2. So $|H| \geq 3r - 2$ by Corollary 1. $\square$

4.5 What stops at $r = 15$

At $r = 15$ the cases are $(u, d, t) = (9, 1, 7)$ and $(11, 5, 5)$. The first is excluded by Proposition 1. The second is open, and both tools lose their grip on it for the same reason.

At $d = 5$ a local graph can be a star, all its edges at one vertex in three colours, with $S_j = 11 = u$. A star forces nothing: an edge through its centre meets every edge of it, so no colour is good and its anchor value is 0. The supply bound of Lemma 13 also fails there, for 4,536 multicoloured graphs.

The swap lemma fails for a similar reason. With $d = 5$ each card of J may carry up to five units of slack, so J may have up to 65 symbols of degree two, with up to 130 incidences with deleted cards. Lemma 8 makes each group need at least 18 of them, 72 in all, which the supply covers. A deleted card can also meet J as a star, all its triples through one card of J , and stars at a common centre always meet.

So Theorem 1 gives only $g(15) \geq 42$. Together with [4] and [P1]–[P4], the bound $g(r) \geq 3r - 2$ is known for $r = 5, 6$ and $r \geq 16$, and open for $7 \leq r \leq 15$.

5. Theorem 3: a gain linear in r

Drop the equality assumption. For the remainder J of degree-four peeling put

$$\ell = r - u - t + 1, \quad h = t + \ell, \quad d = 2r - q + 1,$$

so that $d = u - t - 1 + 2\ell$. Every $x \in U$ has exactly h non-witness symbols, and (1.2) gives $B := q + r + 3 - 4\tau(J) \geq \ell$. With $L = \max(12, d + 8)$, call a one-coloured local graph *large* if its matching number is at least four, and let m be their number.

Lemma 14. *In this general setting:*

(a) $S_x \geq t - 2\ell$ for every $x \in U$, so $S \geq u(t - 2\ell)$ and $u - 1 - S_x \leq d$.

(b) In every F_i , disjoint edges have the same colour, every colour has $2e - s \leq d$, and a vertex carrying E edges in k colours has $E - k \leq d$.

(c) Every F_i that is not large has $S_i \leq L$.

(d) $2m \leq h$.

Proof. (a) The h non-witness symbols of x have degree at most three in J , so each meets at most two further cards, and all $3t$ cards of C must be met: $3t - S_x \leq 2h$.

(b) Two disjoint edges of different colours give two disjoint triples in place of M_i . A card c of M_i meets the other $q - 1$ cards with r symbols of degree at most three, so its total over-coverage is at most $2r - (q - 1) = d$; the witnesses through c cover $2e$ incidences on s vertices. The $u - 1$ witnesses at x reach S_x distinct cards.

(c) If F_i has more than one colour, Lemma 1 with (b) gives $\max(12, u + 2, d + 8)$, and in its star case every edge passes through one vertex p , so (b) at p gives $S_i = E + k \leq d + 2k \leq d + 6$. If F_i has one colour and matching number at most three, a minimum edge cover of its support is a star forest with at most three components, so $e \geq s - 3$ and $2e - s \leq d$ gives $S_i = s \leq d + 6$.

(d) The argument of Lemma 6 uses only the maximality of the triples. A large F_i uses one colour, so its other two points of M_i are met by no witness, and every x reaches these $2m$ points through its h non-witness symbols. A symbol reaching two points of one M_i , together with an edge of F_i avoiding x , gives two triples in place of one; a symbol reaching points of M_i and M_j , together with an edge of F_i avoiding x and an edge of F_j avoiding x and that edge, gives three triples in place of two. $\square$

Theorem 3. *If $u > L$ then*

$$\ell \geq \frac{(u - L)(u - d - 1)}{3u + L}. \quad (5.1)$$

Proof. By Lemma 14, $u(t - 2\ell) \leq S \leq Lt + (u - L)m \leq Lt + (u - L)(t + \ell)/2$, which rearranges to $(u - L)t \leq (5u - L)\ell$; substitute $t = u - d - 1 + 2\ell$. $\square$

Corollary 3. *If the degree-four peeling of H leaves a remainder with $q \geq 2r - 3$, then $|H| \geq \lceil (28r - 67)/9 \rceil$.*

Proof. Here $d \leq 4$ and $L = 12$. For $u > 12$, (5.1) gives $\ell \geq (u - 21)/3$, and with $r = 2u - d - 2 + 3\ell$ this is $\ell \geq (r - 40)/9$. For $u \leq 12$, $r \leq 22 + 3\ell$ gives the same for $r \geq 13$, and for smaller r the bound is negative. So $B \geq (r - 40)/9$, and $|H| = 3r - 3 + E + B + 4C$ (6.1) gives the claim. $\square$

The gain at every remainder size. Write $q = \gamma r$ with $1 \leq \gamma \leq 2$ and $\lambda = \ell/r$. Minimising (5.1) over u , with $r = 2u - d - 2 + 3\ell$, gives $\lambda \geq c(\gamma) - o(1)$, where

$$c(\gamma) = \frac{20 - 4\gamma - \sqrt{(20 - 4\gamma)^2 - 108(\gamma - 1)^2}}{54}. \quad (5.2)$$

If instead $u \leq L$, then $\lambda \geq (\gamma - 1)/3 - o(1)$, which is at least $c(\gamma)$: with $a = \gamma - 1$, c is the smaller root of $27y^2 - (16 - 4a)y + a^2$, whose value at $a/3$ is $\frac{16}{3}a(a - 1) \leq 0$. The gain is $1/9$ at $q = 2r$, about 0.057 at $q = 1.8r$, about 0.019 at $q = 1.5r$ and 0 at $q = r$. For $q < r$ the bound $t \geq 0$ gives $\lambda \geq 1 - \gamma$ directly. Differentiating the quadratic, $c'(\gamma) = (4c + 2(\gamma - 1))/(20 - 4\gamma - 54c)$, which is positive, and below $1/4$ for $1 \leq \gamma \leq 1.81$.

6. Theorems 4 and 5: a coefficient above 3.0534

Stability alone gives nothing near $q = r$. This section adds a second saving, taken from the deleted cards, which is largest exactly there. Together they give Theorem 4.

6.1 Bookkeeping and the clean tail

From here on the peeling always deletes a symbol of **maximum** current degree. For the resulting peeling, with $n = |H|$ and k steps, put

$$E = n - q - 4k, \quad B = q + r + 3 - 4\tau(J), \quad C = k + \tau(J) - r.$$

E and C are non-negative, and so is B , by Theorem 1. Then exactly

$$n = 3r - 3 + E + B + 4C. \quad (6.1)$$

The maximum degree never increases, so after the last step that deletes five or more cards every step deletes exactly four. Call these the *clean* steps, their cards the clean cards, N_c in number, and each step's four cards a clean *group*. Let k_0 be the number of the other steps. Each of them deletes at least five cards, so $E \geq k_0$, and all of the excess E falls in them, so they delete $4k_0 + E$ cards. With (6.1) and $G := B + 4C$ this gives

$$N_c = 3r - 3 - q + G - 4k_0. \quad (6.2)$$

Let H' be J together with the clean cards; it has maximum degree at most four.

Lemma 15 (clean tail). $N_c \leq 3r - q + 1$. Hence $G \leq 4k_0 + 4$.

Proof. Let A be a clean card with group symbol g . Every other symbol v of A has some degree j_v in J , and $j_v + 1 \leq 4$. The card A meets all q cards of J , so $\sum_{v \neq g} j_v \geq q$. It meets the $N_c - 4$ clean cards outside its group, never through g , and v lies in at most $3 - j_v$ clean cards other than A . Adding, $q + N_c - 4 \leq 3(r - 1)$. The second claim is (6.2). $\square$

6.2 Saving group symbols

A *natural cover* of J is made of the t triples of the maximum matching and the witnesses of a pairing of U , plus one symbol for the leftover point when u is odd. It has $t + \lceil u/2 \rceil = \tau(J) + s$ symbols, $s \geq 0$, and $B = \ell + 4s$ for odd u , $B = \ell + 2 + 4s$ for even u .

Lemma 16 (saving). *If a natural cover meets every card of b_4 clean groups and all but one card of b_3 others, then $C + s \geq b_4 + \lfloor b_3/2 \rfloor$.*

Proof. The cover and the k group symbols cover H with $r + C + s$ symbols. Drop the symbols of the b_4 groups. Pair the b_3 others: their two missed cards meet, so one common symbol replaces two group symbols. The result still covers H , so it has at least r symbols. $\square$

So $G \geq \ell + 4(b_4 + \lfloor b_3/2 \rfloor)$, and in particular $G \geq \ell$.

6.3 Traces and the budget

For a clean card A its *trace* L_A is the graph on U with an edge xy whenever the witness w_{xy} lies in A . Let s_A be its number of non-isolated vertices and ν_A its matching number. A minimum edge cover of the non-isolated vertices is a star forest, and one edge from each star is a matching, so $\nu_A \geq s_A - e(L_A)$. Also $\nu_A \leq \lfloor u/2 \rfloor$. Let $M_0 = \sum_A (u - s_A)$ count the missed vertices.

Lemma 17. $M_0 \leq S + 3u\ell$.

Proof. Fix x . A clean card A with x isolated in L_A still meets x , through one of its h non-witness symbols v ; a group symbol lies in no card of J . Such v has $j_v \geq 1$ and lies in at most $4 - j_v$ clean cards. These symbols reach the $3t - S_x$ cards of C that the witnesses miss, so $\sum_v (j_v - 1) \geq 3t - S_x$, and the number of such A is at most $\sum_v (4 - j_v) \leq 3h - (3t - S_x) = S_x + 3\ell$. $\square$

Let $T = \sum_i e(F_i)$ and $D = S - T = \sum_i (S_i - e(F_i))$.

Lemma 18. $D \leq 6t + uh/4$.

Proof. An F_i with more than one colour has $S_i - e(F_i) \leq 6$: the proof of the second bound of Lemma 5 uses only that disjoint edges have the same colour, which Lemma 14(b) gives here. A one-coloured F_i has $S_i - e(F_i) = s - e$, at most its number of components, which is at most its matching number: at most three if it is not large, and at most $\lfloor u/2 \rfloor$ if it is. With m large graphs, $D \leq 6(t - m) + m\lfloor u/2 \rfloor$, and $2m \leq h$ by Lemma 14(d). $\square$

Lemma 19 (budget). *Summed over the clean cards,*

$$\sum_A \left(1 - \frac{2\nu_A}{u}\right) \leq 3\ell + 4 + 4k_0 - G + \frac{h}{2} + \frac{12t}{u}.$$

Proof. By $\nu_A \geq s_A - e(L_A)$, the sum of $u - 2\nu_A$ is at most $2M_0 - N_c u + 2\sum_A e(L_A)$. A witness of degree j in J lies in at most $4 - j$ clean cards. The witnesses of degree three are exactly the edges of the F_i , T in number, so $\sum_A e(L_A) \leq u(u - 1) - T$. With Lemma 17 the sum is at most $2D + 6u\ell + 2u(u - 1) - N_c u$. Divide by u , use Lemma 18, and substitute (6.2) with $q = 3t + u$ and $t = r - u + 1 - \ell$. $\square$

Every term on the left lies in $[0, 1]$.

6.4 Hitting many cards at once

Theorem ([5], Theorem 1.2). *For all sufficiently large N , an N -edge-coloured multigraph in which every colour class is a matching of at least $N + 20N^{15/16}$ edges has a rainbow matching using every colour.*

Lemma 20. *Let N be large and let $\mathcal{A}$ be a set of N clean cards with $\nu_A \geq N + 20N^{15/16}$ for each. Then one natural cover meets every card of $\mathcal{A}$.*

Proof. Give a maximum matching of each L_A the colour A ; a pair that serves two cards becomes two parallel edges. The rainbow matching gives disjoint pairs, one per card, and the witness of the pair coloured A lies in A . Extend to a pairing of U . $\square$

Lemma 21 (allocation). *Fix $0 < \theta < 1/2$ and call a clean card good if $\nu_A \geq \theta u$. Let b be the number of the others. Then $b(1 - 2\theta)$ is at most the right side of Lemma 19, and if any N good cards can be met at once, some natural cover has*

$$b_4 + \lfloor b_3/2 \rfloor \geq \min\left(\frac{N}{6}, \frac{N_c}{4} - \frac{b}{2}\right) - 1.$$

Proof. A card that is not good has $1 - 2\nu_A/u > 1 - 2\theta$; use Lemma 19. For the second claim let g_4 groups have no bad card and g_3 exactly one, so $b \geq g_3 + 2(N_c/4 - g_4 - g_3)$. If $g_4 \geq N/4$, meet all four cards of $\lfloor N/4 \rfloor$ such groups. Otherwise meet all cards of the g_4 groups and three good cards of $\min(g_3, \lfloor (N - 4g_4)/3 \rfloor)$ others. Since $g_3 \geq N_c/2 - b - 2g_4$, this saves at least $g_4 + \frac{1}{2} \min(N_c/2 - b - 2g_4, (N - 4g_4)/3) - 1$, which is at least $\min(N_c/4 - b/2, N/6 + g_4/3) - 1$. $\square$

6.5 The theorem

$$x_* = \frac{65 - 36\sqrt{2}}{138} = 0.10208921 \dots$$

Theorem 4. $\liminf_{r \rightarrow \infty} \frac{g(r)}{r} \geq 3 + x_* = 3.10208921 \dots$

As a decimal coefficient this is 3.1020, rounded down. It improves the constant $(41 - \sqrt{19})/12 \approx 3.0534$ of [2].

Proof. Suppose not. Take $\varepsilon > 0$ and families with $r \rightarrow \infty$ and $n/r \leq 3 + x_* - \varepsilon$, and pass to a subsequence along which the normalised quantities below converge. If J is empty the group symbols cover H , so $n \geq 4r$. Write

$$x = \frac{n}{r} - 3, \quad z = \frac{G}{r}, \quad \lambda = \frac{\ell}{r}, \quad \gamma = \frac{q}{r}, \quad \sigma = \frac{4k_0 - G}{r}, \quad v = \frac{u}{r} = \frac{3 - \gamma - 3\lambda}{2},$$

and $t/r = (\gamma - 1 + \lambda)/2$, $h/r = (\gamma - 1 + 3\lambda)/2$. In the limit:

- $\sigma \geq 0$ (Lemma 15), $x = E/r + z \geq k_0/r + z$, so $x \geq \frac{5}{4}z + \frac{1}{4}\sigma$;
- $z \geq \lambda$ (Lemma 16), and so $\lambda \leq z \leq \frac{4}{5}x < \frac{1}{12}$;
- $\lambda \geq c(\gamma)$ for $1 \leq \gamma \leq 2$ and $\lambda \geq 1 - \gamma$ for $\gamma < 1$ (§5; $\gamma \leq 2$ since $d \geq 0$);
- by (6.2), N_c/r tends to $N := 3 - \gamma - \sigma$;
- by Lemma 19, $\frac{1}{r} \sum_A (1 - 2\nu_A/u)$ has limit at most $M := \sigma + \frac{1}{4}(\gamma - 1 + 15\lambda)$.

Since $\lambda < 1/12$, $\gamma \leq 2$ and $\sigma \geq 0$, we have $v > 0.37$ and $v/N \geq 3/8$. Put

$$K = 2(N - 2M) = 7 - 3\gamma - 15\lambda - 6\sigma.$$

We show $x \geq (15\lambda + 7 - 3\gamma)/24$.

If $K \leq 0$, then $\sigma \geq (7 - 3\gamma - 15\lambda)/6$, and $x \geq \frac{5}{4}\lambda + \frac{1}{4}\sigma$ gives the claim.

If $K > 0$, fix $\theta = K/(20v)$. Since the left side of Lemma 19 is non-negative, $M \geq 0$, so $K \leq 2N$ and $\theta \leq 2N/(20v) \leq 4/15$. Apply Lemma 20 with $N' = \lfloor \theta u - 20(\theta u)^{15/16} \rfloor$, which tends to infinity, and then Lemmas 21 and 16. Dividing by r , and writing $2M = N - K/2$,

$$z - \lambda \geq \min\left(\frac{2\theta v}{3}, N - \frac{2M}{1 - 2\theta}\right) - o(1).$$

The first term equals $K/30$. The second equals

$$\frac{K \left(\frac{1}{2} - \frac{N}{10v} \right)}{1 - \frac{K}{10v}} \geq \frac{7K}{30},$$

since $N/v \leq 8/3$ and the denominator lies in $(0, 1]$. So $z \geq \lambda + K/30$, and

$$x \geq \frac{5}{4} \left(\lambda + \frac{K}{30} \right) + \frac{\sigma}{4} = \frac{15\lambda + 7 - 3\gamma}{24}.$$

In both cases σ has dropped out. The coefficient of λ is positive, so:

- If $1 \leq \gamma \leq 2$: $x \geq f(\gamma) := (15c(\gamma) + 7 - 3\gamma)/24$. The square root in (5.2) is the geometric mean of $20 - 4\gamma \pm \sqrt{108}(\gamma - 1)$, two affine functions positive on $[1, 2]$, so it is concave and c is convex. Hence f is convex, and its minimum is where $c'(\gamma) = 1/5$, namely at $\gamma_* = (21 + 74\sqrt{2})/69 = 1.82104\dots$, with $c(\gamma_*) = (24 - 14\sqrt{2})/69$ and $f(\gamma_*) = x_*$.
- If $\gamma < 1$: $\lambda \geq 1 - \gamma$ gives $x \geq (22 - 18\gamma)/24 \geq 1/6 > x_*$.

Either way $x \geq x_*$, against $x \leq x_* - \varepsilon$. $\square$

At the minimum $K = 0$, that is $\sigma = (10 - 2\sqrt{2})/69$ and $M = N/2$: the bound on the good cards gives nothing there, and about half of the clean cards have traces with small matching number, two in each group. Section 6.6 shows that such cards still carry many witnesses, and uses them.

6.6 The weak traces

Keep the setting and the limits of the proof of Theorem 4, and put

$$\delta = z - \lambda, \quad K_- = \max(0, -K).$$

So $\delta \geq 0$ and $x \geq \frac{5}{4}(\lambda + \delta) + \frac{1}{4}\sigma$.

Lemma 22 (the best θ). *If $K > 0$ then $\delta \geq K/30$. If moreover*

$$\theta_* = \frac{v + 3N - \sqrt{(v + 3N)^2 - 6vK}}{4v} < \frac{1}{2},$$

then

$$\delta \geq R(K) := \frac{v + 3N - \sqrt{(v + 3N)^2 - 6vK}}{6} = \frac{vK}{v + 3N + \sqrt{(v + 3N)^2 - 6vK}} \geq \frac{K}{18}.$$

Proof. The first bound is §6.5. At any fixed $\theta < 1/2$ the proof of §6.5 gives $\delta \geq \min(2\theta v/3, (K/2 - 2N\theta)/(1 - 2\theta)) - o(1)$, since $2M = N - K/2$. The two terms are equal when $4v\theta^2 - (2v + 6N)\theta + 3K/2 = 0$, whose smaller root is θ_* , and there both equal $R(K)$. The last inequality uses $v/N \geq 3/8$. $\square$

With $\lambda = c(\gamma)$ and σ as in §6.5 this already gives $x \geq (15\lambda + 7 - 3\gamma)/24 + K/36$ for $K > 0$, so a value of x near x_* forces K near 0.

Next, a bound on S that does not assume equality in §5. Write $\hat{t} = t/r$ and

$$\bar{s} = \min \left(\hat{t}, \frac{(2-\gamma)\hat{t} + (v-2+\gamma)(\hat{t}+\lambda)/2}{v} \right) \quad (v > 2-\gamma), \quad \bar{s} = \hat{t} \quad (v \leq 2-\gamma).$$

Then $S \leq (\bar{s} + o(1))ur$: a large F_i has $S_i \leq u$, the others have $S_i \leq L$ (Lemma 14(c)), $2m \leq h$, every $S_i \leq u + 4$ (Lemma 1), and $L/r \rightarrow 2 - \gamma$. Put

$$Q = \frac{N}{2} - 3\lambda - \bar{s}.$$

Lemma 23 (weak traces).

$$\delta \geq \min \left(\frac{v}{16}, \frac{v(Q - K_-)_+}{2 + 9v + 9N} \right).$$

Proof. We may assume $\delta < v/16$. Fix $\alpha > 2\delta/v$ with $\alpha < 1/8$ (when $\delta = 0$, any small α , and let $\alpha \rightarrow 0$ at the end). Call a clean card *weak* if $\nu_A < \alpha u$ and *strong* if $\nu_A \geq u/4$.

Groups with three cards that are not weak. Let J_3r be their number. Meeting three such cards in each of $\min(J_3r, \alpha u/3)$ of these groups by Lemma 20, Lemma 16 gives $\delta \geq 2 \min(J_3, \alpha v/3) - o(1)$. As $2\alpha v/3 > \delta$, $J_3 \leq \delta/2 + o(1)$.

Groups of the wrong shape. Call a group *good* if it has two strong cards and two weak ones. A group with at most two cards that are not weak has $\sum \nu_A/u \leq 1 + 2\alpha$ if it is good and at most $\frac{3}{4} + 2\alpha$ if not; a group with three such cards has at most 2. Lemma 19 gives $\frac{1}{r} \sum_A \nu_A/u \geq \frac{N}{4} + \frac{K}{8} - o(1)$. There are $Nr/4$ groups, so the number Zr of groups that are not good satisfies

$$Z \leq 2N\alpha + \frac{5}{2}\delta + \frac{1}{2}K_- + o(1).$$

The witnesses of the weak cards. By Lemma 17 the weak cards of the good groups, $(N/2 - 2Z)r$ of them, miss at most $S + 3u\ell$ vertices in all. Each has $e(L_A) \geq s_A - \nu_A \geq s_A - \alpha u$. So their traces have at least $(Q - 2Z - \alpha N/2 - o(1))ur$ edges in all.

One weak card per good group. Give each good group one colour, carried by the union of the traces of its two weak cards. A colour has at most $2r$ edges, and at most $2(u-1)$ coloured edges meet a vertex, since a witness lies in at most two clean cards. Take a maximal set of disjoint edges of distinct colours, L_w of them. Every edge of an unused colour meets one of their $2L_w$ vertices, so the number of coloured edges, at least half the total above, is at most $2rL_w + 4uL_w$.

The strong cards. Keep $L' = \min(L_w, \lfloor u/32 \rfloor)$ of these edges. Remove their $2L'$ vertices from the traces of the $2L'$ strong cards of the same groups; each keeps a matching of at least $u/4 - 2L' \geq 2L' + 20(2L')^{15/16}$ edges for large r . Lemma 20 meets all of them with disjoint pairs, which together with the L' edges extend to one pairing of U . Its natural cover meets three cards in each of L' groups, and Lemma 16 gives $\delta \geq 2L'/r - o(1)$.

Putting the steps together, with $\alpha \downarrow 2\delta/v$,

$$\delta \geq \min \left(\frac{v}{16}, \frac{v}{2+4v} \left(Q - K_- - \delta \left(5 + \frac{9N}{v} \right) \right)_+ \right),$$

and moving the δ term to the left gives the lemma. $\square$

Theorem 5. $\liminf_{r \rightarrow \infty} \frac{g(r)}{r} \geq 3.1024745$.

Proof. Suppose not, and take the limits of §6.5 with $x < 0.1024745$. Then $1 \leq \gamma \leq 2$ (for $\gamma < 1$, §6.5 gives $x \geq 1/6$), $c(\gamma) \leq \lambda \leq 0.082$ and $0 \leq \sigma \leq 0.41$, and

$$x \geq \frac{5}{4}(\lambda + \max(\delta_1, \delta_2)) + \frac{\sigma}{4},$$

where δ_1 is the bound of Lemma 22 (0 when $K \leq 0$) and δ_2 that of Lemma 23. The script `code/verify_weak_trace_bound.py` shows, by branch and bound over this box with outward-rounded interval arithmetic, that the right side is at least 0.1024745 everywhere. This is a contradiction. $\square$

The numerical minimum of the right side is $0.10247458307\dots$, at $\gamma \approx 1.856983$, $\lambda = c(\gamma) \approx 0.068491$, $\sigma \approx 0.066398$, where $\delta_1 = \delta_2 \approx 0.00020866$. There $K > 0$ and $Q > 0$: neither saving is zero, and the two are in balance. The constant has no closed form that we know of; it is certified to seven decimals, not computed exactly.

The proofs of Theorems 4 and 5 use [5] and give no explicit threshold in r , like [2].

7. Verification

claim	script
Theorem 1 on every configuration with at most six cards, and at low degree up to nine; the sharpness family	<code>code/verify_residual_bound.py</code>
Lemmas 1, 3 and 5, the $d = 0$ structure of §2.3, and the $u = 13$, $d = 5$ step	<code>code/verify_local_graphs.py</code>
Lemmas 10 and 11 on 8, 9 and 10 vertices	<code>code/verify_rainbow_lemmas.py</code> , <code>code/verify_rainbow.c</code>
Lemma 13, the anchor values, and the failure at $d = 5$ reported in §4.5	<code>code/verify_anchor_count.py</code>
every arithmetic step of §2.3, §3.4, §4, §5 and §6, and the constant of [2]	<code>code/verify_p5_arithmetic.py</code>
the algebra of Lemma 19 and Theorem 4: the budget identity, $K = 2(N - 2M)$, the second term of Lemma 21, the reduction of both cases, the root argument for $u \leq L$, convexity of c , $c'(\gamma_*) = 1/5$, $c(\gamma_*)$, σ_* and the constant; the optimal θ and $R(K) \geq K/18$ of Lemma 22; the counts of Lemma 23	<code>code/verify_rainbow_constant.py</code>
Theorem 5: the lower bound 0.1024745 over the whole domain, by branch and bound with outward-rounded interval arithmetic	<code>code/verify_weak_trace_bound.py</code>

The runs are recorded in `results/p5_checks.txt`. These are finite checks of the finite lemmas. The steps that are not finite, namely (2.1), (2.2), Lemma 4, the inequality $M \leq S$, the swap lemma,

the augmentation arguments of Lemmas 6, 9, 12 and 14, and Lemmas 15 to 23, have been read, Lemmas 15 to 21 in a separate review session, but not independently reviewed. Theorems 4 and 5 also rest on the published theorem of [5].

This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository README sets out the division of labour and the open obligations in full.

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